How to solve these exercises

A plain-language walkthrough, written so that a grade 9 student can follow it. It has two parts. The toolkit comes first: every piece of maths the exercises rely on, each with a small example. The walkthrough of the exercises comes after, and it points back to the toolkit by number (T1, T2, and so on) whenever it uses a tool. The finished proofs are still in `solutions.tex` if you want to check your work.

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Toolkit: what you need to know first

You do not have to read this whole part in one go. Skim the headings, read the tools you have not met before, and come back when the walkthrough mentions one. This table shows which tools each exercise uses.

Exercise	Tools it uses
0.1	T1, T2, T3, T6, T7, T8, T9, T10
0.2	T2, T3, T4, T8
0.3	T2, T5, T6, T11, T16
1.1	T4, T5, T12
1.2	T4, T6, T12, T13
1.3	T4, T6, T8, T13, T14
1.4	T4, T6, T9
1.5	T4, T8, T16
1.9 and 1.10	T5, T15

T1. Functions, and what n means

A function is a rule that turns one number into another. $f(n) = 3n$ is the rule "triple it." So $f(2) = 6$ and $f(10) = 30$. The brackets do not mean multiplication here; $f(n)$ is read "f of n."

In these exercises n is always a whole number that counts something, such as how many items are in a list. The names f and g are just labels for two rules being compared.

A few words that come up throughout:

- An **integer** is a whole number: $\ldots, -2, -1, 0, 1, 2, \ldots$
- A **constant** is a fixed number that does not change when n changes, like 100 or $\frac{1}{2}$.
- **Terms** are things being added. **Factors** are things being multiplied. In $3n+5$ the terms are $3n$ and 5; in $3n$ the factors are 3 and n.
- A **ratio** is one quantity divided by another, written as a fraction.
- An **identity** is an equation that is true for every value you put in, like $2(x+1) = 2x+2$.

T2. Powers and the exponent rules

2^5 means five 2s multiplied together: $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 32$. The 2 is the base and the 5 is the exponent. (A dot between numbers means "times.")

Rule	In words	Example
$a^m \cdot a^k = a^{m+k}$	multiplying adds exponents	$2^3 \cdot 2^2 = 8 \cdot 4 = 32 = 2^5$
$\dfrac{a^m}{a^k} = a^{m-k}$	dividing subtracts exponents	$\dfrac{2^5}{2^2} = \dfrac{32}{4} = 8 = 2^3$
$(a^m)^k = a^{mk}$	a power of a power multiplies	$(2^3)^2 = 8^2 = 64 = 2^6$
$a^0 = 1$	exponent zero gives 1	$7^0 = 1$
$a^{-k} = \dfrac{1}{a^k}$	a negative exponent means "one over"	$2^{-3} = \dfrac{1}{8}$
$a^{1/2} = \sqrt{a}$	exponent one half is a square root	$9^{1/2} = 3$
$\left(\dfrac{a}{b}\right)^k = \dfrac{a^k}{b^k}$	a power goes onto top and bottom	$\left(\dfrac{2}{3}\right)^2 = \dfrac{4}{9}$

Three consequences get used a lot:

- One extra in the exponent is one extra factor: $2^{n+1} = 2 \cdot 2^n$.
- An exponent can be a fraction or a decimal. $n^{2/3}$ and $n^{0.01}$ are ordinary powers. As long as the exponent is positive, the power still grows when n grows, just slowly.
- Powers of a number bigger than 1 grow: $2, 4, 8, 16, \ldots$ Powers of a number between 0 and 1 shrink: $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \ldots$

T3. Working with fractions

- **Same thing to top and bottom.** Dividing the top and bottom by the same number does not change a fraction: $\frac{6}{8} = \frac{3}{4}$. This also works with n: $\frac{3n}{n} = 3$.
- **Splitting the top.** $\frac{n-100}{n} = \frac{n}{n} - \frac{100}{n} = 1 - \frac{100}{n}$.
- **Dividing by a fraction.** Flip it and multiply: $\frac{1}{2} \div \frac{3}{4} = \frac{1}{2} \cdot \frac{4}{3} = \frac{4}{6} = \frac{2}{3}$.

- **Common denominators.** To add or subtract, make the bottoms match first: $\frac{1}{2} - \frac{2}{3} = \frac{3}{6} - \frac{4}{6} = -\frac{1}{6}$.
- **Bigger bottom, smaller fraction.** $\frac{1}{8}$ is less than $\frac{1}{4}$. A bigger top makes a bigger fraction.

T4. Inequalities

$a < b$ means a is less than b. $a \leq b$ means less than or equal. $>$ and $\geq$ are the same ideas the other way round. $\approx$ means "approximately equal," and $\neq$ means "not equal."

- **Chains.** If $a \leq b$ and $b \leq c$, then $a \leq c$. Proofs here are often long chains like this, each link making the right side a little bigger and simpler.
- **Same thing to both sides.** You may add or subtract the same number on both sides, or multiply or divide both sides by the same positive number. The inequality keeps its direction.
- **Bounding a sum.** If you add up k numbers and none is bigger than M, the total is at most $k \cdot M$. If none is smaller than m, the total is at least $k \cdot m$. The same idea works for products: k factors that are each at most M multiply to at most M^k.
- **Upper and lower bounds.** An upper bound is something you know the answer is below. A lower bound is something you know it is above.

T5. Expanding and factoring

Expanding means multiplying brackets out. Factoring is the reverse.

- One bracket: $3(b-1) = 3b - 3$.
- Two brackets: multiply each term in the first by each term in the second.

$$(a+b)(c+d) = ac + ad + bc + bd.$$

- Squaring a bracket is a special case: $(a+b)^2 = a^2 + 2ab + b^2$.
- Factoring out something shared: $aN + bN = (a+b)N$.
- Factoring a quadratic: $b^2 - 3b + 2 = (b-1)(b-2)$. Check by expanding the right side: $b^2 - 2b - b + 2$.

T6. Logarithms

A logarithm answers the question "what exponent do I need?"

$$\log_2 8 = 3 \quad \text{because} \quad 2^3 = 8.$$

The small 2 is the base. More examples: $\log_2 32 = 5$ because $2^5 = 32$, and $\log_{10} 1000 = 3$ because $10^3 = 1000$. Answers do not have to be whole numbers: $\log_2 10 \approx 3.322$, because $2^{3.322} \approx 10$.

Logs and powers undo each other:

$$\log_b (b^x) = x \qquad \text{and} \qquad b^{\log_b x} = x.$$

Rule	In words	Example with base 2
$\log(ab) = \log a + \log b$	times becomes plus	$\log_2(4 \cdot 8) = 2 + 3 = 5$
$\log \dfrac{a}{b} = \log a - \log b$	divide becomes minus	$\log_2 \dfrac{8}{2} = 3 - 1 = 2$
$\log\left(a^k\right) = k \log a$	an exponent comes out the front	$\log_2\left(4^3\right) = 3 \cdot 2 = 6$

Things to know about the way logs are written here:

- $\log n$ with no base written means "pick any base bigger than 1; it will not matter." The reason is the change-of-base rule: $\log_2 n \approx 3.322 \cdot \log_{10} n$. Switching base only multiplies by a fixed number.
- ln is a log with a special base, $e \approx 2.718$.
- $(\log n)^2$ and $\log^2 n$ both mean "take the log, then square the answer." That is different from $\log(n^2)$, which means "square first, then take the log."
- $\log(\log n)$ is a log of a log. If $n = 65{,}536$ then $\log_2 n = 16$ and $\log_2(\log_2 n) = \log_2 16 = 4$.
- Logs grow very slowly. $\log_2$ of a thousand is about 10. $\log_2$ of a million is about 20. $\log_2$ of a billion is about 30.

T7. "Goes to": what happens when n gets huge

The arrow $\to$ is read "goes to" or "approaches." It describes where an expression is heading as n grows without stopping. The symbol ∞ means infinity. It is not a number; "$n \to \infty$" just means "n keeps growing forever."

n	$100/n$	$1 - 100/n$
1,000	0.1	0.9
100,000	0.001	0.999
10,000,000	0.00001	0.99999

So $100/n \to 0$ and $1 - 100/n \to 1$. The value it is heading for is called the limit. Three facts cover every limit in this assignment:

- A fixed number divided by something that grows forever goes to 0.
- Something that grows forever, divided by a fixed number, still grows forever.
- Once you know where each piece is heading, you can swap those values in. That is how $\frac{1-100/n}{1-200/n}$ goes to $\frac{1-0}{1-0} = 1$.

T8. Sums written with dots and with Σ

Three dots mean "carry on with the same pattern." $1+2+3+\cdots+n$ is every whole number from 1 to n added up.

The symbol Σ (capital sigma) is shorthand for the same thing:

$$\sum_{i=1}^{4} i^2 = 1^2 + 2^2 + 3^2 + 4^2 = 1 + 4 + 9 + 16 = 30.$$

Read it as: let i be 1, then 2, then 3, up to the number on top; work out the expression each time; add up the results. Dots are also used for products, as in $1 \cdot 2 \cdot 3 \cdots n$.

T9. Factorials

$n!$ is read "n factorial." It means every whole number from 1 to n multiplied together:

$$5! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 = 120.$$

Factorials get enormous very quickly: 10! is already 3,628,800.

T10. How fast different formulas grow

The whole assignment is about comparing growth. Seeing real numbers helps.

n	$\log_2 n$	$\sqrt{n}$	n	n^2	2^n	$n!$
10	3.3	3.2	10	100	1,024	3,628,800
20	4.3	4.5	20	400	1,048,576	about 2.4×10^{18}
100	6.6	10	100	10,000	about 1.3×10^{30}	about 9.3×10^{157}

Each column eventually leaves the one before it far behind. Notice the word "eventually": at $n = 10$ the log is still slightly ahead of the square root. By $n = 100$ it has lost, and it never catches up.

T11. Square roots and the number φ

$\sqrt{5}$ is the positive number whose square is 5. It is about 2.236, and its decimals never end or repeat. The useful fact is simply $\sqrt{5} \cdot \sqrt{5} = 5$.

φ is the Greek letter phi. In exercise 0.3 it is the name of one particular number, called the golden ratio:

$$\varphi = \frac{1+\sqrt{5}}{2} \approx \frac{1+2.236}{2} = 1.618.$$

T12. Writing numbers in other bases

Ordinary numbers are in base 10: each place is worth ten times the place to its right. 345 means $3 \cdot 100 + 4 \cdot 10 + 5 \cdot 1$, and the digits run from 0 to 9.

In base b, each place is worth b times the one to its right, and the digits run from 0 to $b-1$. Base 2 is called binary, its digits are just 0 and 1, and its places are worth $1, 2, 4, 8, 16, \ldots$ A binary digit is called a bit. A small subscript shows the base:

$$1101_2 = 1 \cdot 8 + 1 \cdot 4 + 0 \cdot 2 + 1 \cdot 1 = 13.$$

Two facts about length:

- In base b, the smallest three-digit number is $100_b = b^2$, so the largest two-digit number is $b^2 - 1$. In base 10 those are 100 and 99.
- A number with d decimal digits is at least 10^{d-1} and less than 10^d. For $d = 3$, that is from 100 up to 999. This is why the number of digits is tied to a log: the number of digits of N in base b is $\log_b N$ rounded down, plus 1. For $N = 345$, $\log_{10} 345 \approx 2.54$, which rounds down to 2, and $2 + 1 = 3$ digits.

T13. Rounding down and rounding up

$\lfloor x \rfloor$ is called the floor of x. It means round down to a whole number: $\lfloor 3.7 \rfloor = 3$.

$\lceil x \rceil$ is called the ceiling. It means round up: $\lceil 3.2 \rceil = 4$. A whole number stays as it is: $\lceil 3 \rceil = 3$.

Rounding down never increases a number, so $\lfloor x \rfloor \leq x$.

T14. Trees

In computer science a tree is a diagram like a family tree drawn upside down. The dots are called nodes. The one at the top is the root. The nodes directly below a node are its children. A d-ary tree is one where no node has more than d children; with $d = 2$ it is called a binary tree.

The depth of a node is how many steps it is below the root, so the root has depth 0. The depth of the tree is the depth of its lowest node.

T15. Divisibility and remainders

"5 divides 15" means 15 is a whole number of 5s: $15 = 3 \cdot 5$. You can also say 15 is a multiple of 5. In general, "N divides m" means $m = kN$ for some whole number k (which may be zero or negative).

When a division does not come out evenly, there is a remainder: $17 \div 5$ is 3 remainder 2, because $17 = 3 \cdot 5 + 2$.

The symbol $\equiv$ compares remainders. $17 \equiv 2 \pmod 5$ is read "17 is congruent to 2, mod 5," and it means 17 and 2 leave the same remainder when divided by 5. A clock is the everyday version: 15 o'clock and 3 o'clock are the same position, because $15 \equiv 3 \pmod{12}$.

A dash on a letter, as in x', is read "x prime." It is just a second name, used for a number that is related to x.

T16. Two ways of proving things

A proof is an argument that shows a statement is always true, not only for the examples you tried.

Trapping. To show a messy quantity grows like something simple, find a simple thing that is always above it and a simple thing that is always below it. If the two simple things grow at the same speed, the messy one is squeezed between them and must grow at that speed too.

Induction. This proves a statement for every whole number from some starting point. Picture a line of dominoes. If you knock over the first one, and every falling domino knocks over the next, then they all fall. In a proof:

1. The base case is the first domino: check the statement directly for the starting number.
2. The inductive step is "each domino knocks over the next": assume the statement is true for the earlier numbers and show it must then be true for the next one.

T17. The Greek letters, in one place

Symbol	Name	What it means here
O	big O	grows no faster than
Ω	big omega	grows at least as fast as
Θ	big theta	grows at the same speed as
Σ	sigma	add up a list of terms (T8)
φ	phi	the number 1.618… (T11)
γ	gamma	the number 0.577…, mentioned in 1.5 but not needed

The first three are explained in the next section.

First, the words these problems use

Most of this assignment is one question: when n gets huge, which formula grows faster?

Think of n as the size of a problem. $n = 10$ is a short list. $n = 1{,}000{,}000$ is a long one. We do not care what happens at $n = 3$. We care what happens eventually.

SAY IT IN ENGLISH

$f = O(g)$ means "f does not grow faster than g." g is a ceiling. f might be slower, or they might be about the same speed.

$f = \Omega(g)$ means "f grows at least as fast as g." g is a floor.

$f = \Theta(g)$ means both. Same speed, up to a fixed multiplier like 2 or 100. The symbol is a capital theta, and people say "theta."

The practical test is to divide them and see where the fraction is heading as n gets huge (T7).

$$\frac{f(n)}{g(n)}$$

- If the fraction settles near a normal positive number, like 5 or $1/10$, they are the same speed: Θ.
- If the fraction shrinks toward 0, then f is slower: O only.
- If the fraction blows up toward infinity, then f is faster: Ω only.

Example: $f(n) = 3n$ and $g(n) = n$. The fraction is $3n/n = 3$. Same speed, even though one is always triple the other. A fixed multiplier does not count as "faster" here.

WHAT "FASTER" DOES NOT MEAN

$100n$ is bigger than n for every n, but they are still the same kind of growth. Both are straight lines. n^2 really is faster than $100n$, because $n^2/100n = n/100$ goes to infinity. At $n = 10$, the square is still smaller. At $n = 10{,}000$, the square has won forever. "Eventually" is the whole game.

One ladder, from slow to fast. You can trust it for the rest of this guide, and the table in T10 shows it with real numbers:

$$\text{constants} \quad < \quad \log n \quad < \quad \sqrt{n} \quad < \quad n \quad < \quad n^2 \quad < \quad 2^n \quad < \quad n!$$

Any power of a log loses to any positive power of n. So $(\log n)^{100}$ still loses to $\sqrt{n}$. It just takes longer to lose.

Changing the base of a log only multiplies by a constant. $\log_{10} n$ and $\log_2 n$ are the same speed. Pick whichever base makes the algebra nicer.

0.1 Which function grows faster?

Tools used: T1, T2, T3, T6, T7, T8, T9 and T10.

For each pair, simplify f/g until you can see the limit. Do not stare at the original formulas and guess.

Ignore makeup

(a) $n - 100$ versus $n - 200$. Write the fraction, then divide the top and bottom by n:

$$\frac{n-100}{n-200} = \frac{(n-100)/n}{(n-200)/n} = \frac{1-100/n}{1-200/n}.$$

As n gets huge, $100/n$ and $200/n$ both shrink to 0, so the fraction gets closer and closer to

$$\frac{1-0}{1-0} = 1.$$

Same speed. Answer: Θ.

The same cleanup handles anything that is "a number times n, plus slower junk."

(c) $100n + \log n$ over $n + (\log n)^2$. Divide top and bottom by n:

$$\frac{100n + \log n}{n + (\log n)^2} = \frac{100 + (\log n)/n}{1 + (\log n)^2/n}.$$

Logs lose to n on the ladder, so $(\log n)/n$ and $(\log n)^2/n$ both shrink to 0. The fraction goes to

$$\frac{100 + 0}{1 + 0} = 100.$$

A normal positive number, so Θ.

(d), (e), (f) are logs wearing a disguise. Two log rules do all the work: $\log(ab) = \log a + \log b$ and $\log(a^k) = k \log a$.

(d) $n \log n$ over $10n \log(10n)$. Cancel the n, split the log, then divide top and bottom by $\log n$:

$$\begin{aligned}
\frac{n \log n}{10n \log(10n)} &= \frac{\log n}{10 \log(10n)} && \text{cancel } n \\
&= \frac{\log n}{10(\log 10 + \log n)} && \text{split the log} \\
&= \frac{1}{10\left(\dfrac{\log 10}{\log n} + 1\right)} && \text{divide by } \log n \\
&\to \frac{1}{10(0+1)} = \frac{1}{10}.
\end{aligned}$$

Θ.

(e) $\log(2n)$ over $\log(3n)$. Split both logs, then divide top and bottom by $\log n$:

$$\frac{\log(2n)}{\log(3n)} = \frac{\log 2 + \log n}{\log 3 + \log n} = \frac{\dfrac{\log 2}{\log n} + 1}{\dfrac{\log 3}{\log n} + 1} \to \frac{0+1}{0+1} = 1.$$

Θ.

(f) $10 \log n$ over $\log(n^2)$. Pull the exponent out of the log, then cancel:

$$\frac{10 \log n}{\log(n^2)} = \frac{10 \log n}{2 \log n} = \frac{10}{2} = 5.$$

Exactly 5 for every n. Θ.

(n) is the same idea with exponents. Since $2^{n+1} = 2 \cdot 2^n$,

$$\frac{2^n}{2^{n+1}} = \frac{2^n}{2 \cdot 2^n} = \frac{1}{2}.$$

Θ.

Subtract the exponents

(b) $n^{1/2}$ over $n^{2/3}$. Dividing powers of the same base subtracts the exponents:

$$\frac{n^{1/2}}{n^{2/3}} = n^{1/2-2/3} = n^{3/6-4/6} = n^{-1/6} = \frac{1}{n^{1/6}}.$$

The bottom grows without limit, so the fraction goes to 0. The first function is slower. O only.

If the exponent you get is positive, the first function is faster. If it is zero, they match.

Logs lose to powers of n

Cancel the copies of n that appear on both sides, then use the ladder.

(g) $n^{1.01}$ over $n \log^2 n$:

$$\frac{n^{1.01}}{n \log^2 n} = \frac{n^{1.01-1}}{\log^2 n} = \frac{n^{0.01}}{\log^2 n}.$$

A power of n, even $n^{0.01}$, beats any power of $\log n$. The fraction goes to infinity. Ω only.

(h) $n^2/\log n$ over $n(\log n)^2$. Dividing by something is multiplying by its flip:

$$\frac{n^2/\log n}{n(\log n)^2} = \frac{n^2}{\log n} \cdot \frac{1}{n(\log n)^2} = \frac{n^2}{n(\log n)^3} = \frac{n}{(\log n)^3}.$$

A power of n over a power of a log. Goes to infinity. Ω only.

(i) $n^{0.1}/(\log n)^{10}$ is already in that form. Goes to infinity. Ω only.

(k) $\sqrt{n}/(\log n)^3 = n^{1/2}/(\log n)^3$. Same form again. Goes to infinity. Ω only.

You do not need the exact limit. You only need to know it is infinity or zero.

Turn weird exponents into powers of n

ONE IDENTITY WORTH MEMORIZING

$$a^{\log_b n} = n^{\log_b a}.$$

In words: "a raised to a log of n" is just n raised to a constant. The constant is the log of a.

Where it comes from: any number a can be written as $b^{\log_b a}$, because that is what a log means. Then

$$\begin{aligned}
a^{\log_b n} &= \left(b^{\log_b a}\right)^{\log_b n} && \text{rewrite } a \\
&= b^{\log_b a \cdot \log_b n} && \text{power of a power multiplies} \\
&= \left(b^{\log_b n}\right)^{\log_b a} && \text{regroup the product} \\
&= n^{\log_b a} && \text{because } b^{\log_b n} = n.
\end{aligned}$$

(l) $n^{1/2}$ versus $5^{\log_2 n}$. Use the identity with $a = 5$ and $b = 2$:

$$5^{\log_2 n} = n^{\log_2 5} \approx n^{2.32}.$$

Now both sides are powers of n, so subtract the exponents:

$$\frac{n^{1/2}}{n^{\log_2 5}} \approx n^{0.5-2.32} = n^{-1.82} = \frac{1}{n^{1.82}} \to 0.$$

O only.

(m) $n \cdot 2^n$ versus 3^n. Group the two exponentials together:

$$\frac{n \cdot 2^n}{3^n} = n \cdot \frac{2^n}{3^n} = n\left(\frac{2}{3}\right)^n.$$

The fraction $2/3$ is less than 1, so $(2/3)^n$ shrinks very fast. Multiplying by n cannot save it. At $n = 10$ it is about 0.17; at $n = 50$ it is about 0.00000008. It goes to 0. O only.

(o) $n!$ versus 2^n. Both are products of n numbers, so pair them up:

$$\frac{n!}{2^n} = \frac{1 \cdot 2 \cdot 3 \cdot 4 \cdots n}{2 \cdot 2 \cdot 2 \cdot 2 \cdots 2} = \frac{1}{2} \cdot \frac{2}{2} \cdot \frac{3}{2} \cdot \frac{4}{2} \cdots \frac{n}{2}.$$

After the first two factors, every new factor is bigger than 1, and they keep getting bigger. The product runs off to infinity. $n!$ wins. Ω only.

When both sides look scary, take a log

If you only need to know which one wins, look at the log of the fraction. The rule $\log(f/g) = \log f - \log g$ turns the division into a subtraction. If that difference goes to $+\infty$, so does f/g. If it goes to $-\infty$, then f/g goes to 0.

To keep the algebra short, write $L = \log_2 n$. Then $n = 2^L$, and L gets huge when n does.

(j) $(\log n)^{\log n}$ versus $n/\log n$. In terms of L, the two functions are $f = L^L$ and $g = 2^L/L$. Take $\log_2$ of each:

$$\log_2 f = \log_2\left(L^L\right) = L \log_2 L,$$
$$\log_2 g = \log_2\left(\frac{2^L}{L}\right) = \log_2\left(2^L\right) - \log_2 L = L - \log_2 L.$$

Subtract:

$$\log_2 f - \log_2 g = L \log_2 L - L + \log_2 L = L(\log_2 L - 1) + \log_2 L.$$

Both pieces grow without limit, so the difference goes to $+\infty$. The first function wins. Ω only.

(p) That same $(\log n)^{\log n}$ versus $2^{(\log_2 n)^2}$. Now $f = L^L$ again and $g = 2^{L^2}$:

$$\log_2 f = L \log_2 L,$$
$$\log_2 g = \log_2\left(2^{L^2}\right) = L^2.$$

Subtract:

$$\log_2 f - \log_2 g = L \log_2 L - L^2 = L(\log_2 L - L).$$

$\log_2 L$ is far smaller than L, so the bracket is a large negative number, and the difference goes to $-\infty$. The fraction goes to 0. O only.

Same left-hand function, two different opponents, two different answers. Always compare the pair in front of you.

A sum of powers

(q) $\sum_{i=1}^{n} i^k$ means $1^k + 2^k + \cdots + n^k$, with k a fixed number like 2 or 5.

Upper bound. There are n terms, and none is bigger than n^k:

$$1^k + 2^k + \cdots + n^k \le \underbrace{n^k + n^k + \cdots + n^k}_{n \text{ terms}} = n \cdot n^k = n^{k+1}.$$

Lower bound. Keep only the second half of the terms. There are about $n/2$ of them, and each is at least $(n/2)^k$:

$$1^k + 2^k + \cdots + n^k \ge \frac{n}{2} \cdot \left(\frac{n}{2}\right)^k = \left(\frac{n}{2}\right)^{k+1} = \frac{n^{k+1}}{2^{k+1}}.$$

Now divide everything by n^{k+1}:

$$\frac{1}{2^{k+1}} \le \frac{1^k + 2^k + \cdots + n^k}{n^{k+1}} \le 1.$$

The fraction is trapped between two fixed positive numbers. Same speed as n^{k+1}. Θ.

ANSWER KEY FOR 0.1
Same speed, Θ: (a), (c), (d), (e), (f), (n), (q).
Left side slower, O only: (b), (l), (m), (p).
Left side faster, Ω only: (g), (h), (i), (j), (k), (o).

0.2 Adding $1 + c + c^2 + \cdots + c^n$

Tools used: T2, T3, T4 and T8.

Each term is the previous term multiplied by the same number c. The whole exercise is: which term matters?

First, a shortcut formula for the sum

Call the sum S. Multiply it by c and line the two up:

$$\begin{aligned} S &= 1 + c + c^2 + \cdots + c^n, \\ cS &= \quad c + c^2 + \cdots + c^n + c^{n+1}. \end{aligned}$$

Subtract the first line from the second. Every term in the middle appears in both lines and cancels, leaving only the two ends:

$$cS - S = c^{n+1} - 1.$$

Factor out S on the left and divide by $c-1$ (allowed whenever $c \neq 1$):

$$S(c-1) = c^{n+1} - 1 \qquad \Longrightarrow \qquad S = \frac{c^{n+1}-1}{c-1}.$$

Check it with $c = 2$ and $n = 4$: $(2^5 - 1)/(2 - 1) = 31$, and $1 + 2 + 4 + 8 + 16 = 31$.

The three cases

If $c = 1$. The formula does not apply (it would divide by 0), but you do not need it. You are adding 1, a total of $n + 1$ times:

$$S = \underbrace{1 + 1 + \cdots + 1}_{n+1 \text{ terms}} = n + 1.$$

Divide by n to compare: $(n + 1)/n = 1 + 1/n \to 1$. Same speed as n. $\Theta(n)$.

If $0 < c < 1$. The terms shrink. Try $c = 1/2$:

$$1 + \tfrac{1}{2} + \tfrac{1}{4} + \tfrac{1}{8} + \cdots$$

You can keep adding forever and never reach 2. So a finite piece of the sum is stuck between 1 and 2. Stuck between two fixed numbers means $\Theta(1)$: it does not grow.

For any c between 0 and 1, flip the signs on the top and bottom of the formula so both are positive:

$$S = \frac{c^{n+1}-1}{c-1} = \frac{1-c^{n+1}}{1-c}.$$

The top, $1 - c^{n+1}$, is less than 1, because you are subtracting a positive number from 1. So

$$1 \leq S < \frac{1}{1-c}.$$

The left side holds because the first term of the sum is already 1. Both ends are fixed numbers that do not depend on n. $\Theta(1)$.

If $c > 1$. The terms grow, so the last term is the boss. For $c = 2$:

$$1 + 2 + 4 + 8 + 16 = 31,$$

and the last term is 16. The whole sum is less than twice the last term.

For any $c > 1$, the sum is at least its last term, c^n. For the other side, drop the -1 from the top of the formula, which can only make it bigger, then peel one c off c^{n+1}:

$$S = \frac{c^{n+1}-1}{c-1} < \frac{c^{n+1}}{c-1} = \frac{c \cdot c^n}{c-1} = \frac{c}{c-1} \cdot c^n.$$

Put together:

$$c^n \leq S < \frac{c}{c-1} \cdot c^n.$$

The sum is trapped between 1 times the last term and $\frac{c}{c-1}$ times the last term. (For $c = 2$ that multiplier is 2, matching $31 < 2 \cdot 16$.) So the sum is $\Theta(c^n)$.

THE SENTENCE TO REMEMBER

If the terms shrink, the sum is about the first term. If the terms grow, the sum is about the last term. If the terms stay the same, the sum is about how many terms there are.

0.3 Fibonacci grows like an exponential

Tools used: T2, T5, T6, T11 and T16.

The sequence is $0, 1, 1, 2, 3, 5, 8, 13, 21, \ldots$. Each number is the sum of the two before it. Call the nth one F_n, starting at $F_0 = 0$.

These proofs use induction (T16). You can also think of it as a relay race:

1. Check the claim by hand for the first one or two values. Those are the base cases.
2. Assume it is already true for the earlier values.
3. Use the rule $F_n = F_{n-1} + F_{n-2}$ to show the next one is true.

Because Fibonacci looks back two steps, you always need two base cases.

(a) Show $F_n \geq 2^{n/2}$ once $n \geq 6$

$2^{n/2}$ is about 1.414^n. The problem claims Fibonacci grows at least that fast, starting at $n = 6$.

Base cases, just compute:

- $F_6 = 8$, and $2^{6/2} = 8$. Equal, so "at least" holds.
- $F_7 = 13$, and $2^{3.5} \approx 11.3$. 13 is bigger.

Now assume it is true for the two previous numbers, and add them:

$$F_n = F_{n-1} + F_{n-2} \geq 2^{(n-1)/2} + 2^{(n-2)/2}.$$

To compare this with $2^{n/2}$, split each exponent so that $n/2$ stands alone:

$$2^{(n-1)/2} = 2^{n/2-1/2} = 2^{n/2} \cdot 2^{-1/2} = 2^{n/2} \cdot \frac{1}{\sqrt{2}},$$
$$2^{(n-2)/2} = 2^{n/2-1} = 2^{n/2} \cdot 2^{-1} = 2^{n/2} \cdot \frac{1}{2}.$$

Add them and factor out $2^{n/2}$:

$$F_n \geq 2^{n/2} \left(\frac{1}{\sqrt{2}} + \frac{1}{2} \right) \approx 2^{n/2}(0.707 + 0.5) = 2^{n/2} \cdot 1.207.$$

The bracket is more than 1, so $F_n \geq 2^{n/2}$. The relay continues.

(b) Find some $c < 1$ so that $F_n \leq 2^{cn}$

2^{cn} with $c < 1$ grows slower than 2^n. You need an exponential that is still fast enough to stay above Fibonacci.

The clean choice comes from the number

$$\varphi = \frac{1+\sqrt{5}}{2} \approx 1.618.$$

It is less than 2, so $c = \log_2 \varphi \approx 0.694$ is less than 1. And $2^{cn} = \varphi^n$. So the claim you actually prove is $F_n \le \varphi^n$.

Why this number? Because it satisfies $\varphi^2 = \varphi + 1$. That is the same shape as the Fibonacci rule. You can check it by squaring:

$$\varphi^2 = \left(\frac{1+\sqrt{5}}{2}\right)^2 = \frac{1+2\sqrt{5}+5}{4} = \frac{6+2\sqrt{5}}{4} = \frac{3+\sqrt{5}}{2},$$
$$\varphi + 1 = \frac{1+\sqrt{5}}{2} + \frac{2}{2} = \frac{3+\sqrt{5}}{2}.$$

The two match.

Base cases, $n = 0$ and $n = 1$: $F_0 = 0 \le 1 = \varphi^0$ and $F_1 = 1 \le 1.618 = \varphi^1$.

Now assume the claim for the two previous numbers and add them:

$$\begin{aligned}
F_n = F_{n-1} + F_{n-2} &\le \varphi^{n-1} + \varphi^{n-2} && \text{the assumption} \\
&= \varphi^{n-2} \cdot \varphi + \varphi^{n-2} \cdot 1 && \text{split off } \varphi^{n-2} \\
&= \varphi^{n-2}(\varphi + 1) && \text{factor} \\
&= \varphi^{n-2} \cdot \varphi^2 && \text{because } \varphi + 1 = \varphi^2 \\
&= \varphi^n && \text{add the exponents.}
\end{aligned}$$

The algebra closes exactly. Any bigger c that is still under 1 would also work. The problem only asks you to find one.

(c) What is the biggest such c?

Run the same induction the other way. The claim is $F_n \ge \varphi^{n-2}$ for $n \ge 2$.

Base cases: $F_2 = 1 = \varphi^0$ and $F_3 = 2 \ge 1.618 = \varphi^1$.

The step uses exactly the same factoring as part (b), with each exponent two lower:

$$\begin{aligned}
F_n = F_{n-1} + F_{n-2} &\ge \varphi^{n-3} + \varphi^{n-4} \\
&= \varphi^{n-4}(\varphi + 1) \\
&= \varphi^{n-4} \cdot \varphi^2 \\
&= \varphi^{n-2}.
\end{aligned}$$

Now rewrite the result as a fixed fraction of φ^n:

$$F_n \ge \varphi^{n-2} = \frac{\varphi^n}{\varphi^2} \approx \frac{\varphi^n}{2.618} \approx 0.38\,\varphi^n.$$

Fibonacci is at least 38% of φ^n, forever. Since $\varphi^n = 2^{cn}$ with $c = \log_2 \varphi \approx 0.694$, that value of c is allowed.

It is the biggest. Suppose you try a bigger base $B > \varphi$ in place of φ. Part (b) says $F_n \le \varphi^n$, so

$$\frac{F_n}{B^n} \le \frac{\varphi^n}{B^n} = \left(\frac{\varphi}{B}\right)^n.$$

The fraction φ/B is less than 1, so its powers shrink to 0, just like $(2/3)^n$ in exercise 0.1(m). Fibonacci becomes a vanishing fraction of B^n, and "at least as fast" is false.

A CONSTANT IN FRONT IS ALLOWED

Ω lets you multiply by a fixed number, and it lets the inequality start late. It does not let you increase the exponent. $\varphi^n/10$ still counts. $(1.7)^n$ does not, because $1.7 > 1.618$.

1.1 and 1.2 How long is a number?

Tools used: T4, T5, T6, T12 and T13.

1.1 Adding three digits

In base 10, a digit is 0 through 9. Three of them add up to at most 27, which is still only two digits. The problem asks you to see that this happens in every base $b \ge 2$, not just base 10.

In base b, a digit is an integer from 0 to $b-1$. The worst sum is $3(b-1)$. A number needs three digits only if it is at least b^2. In base 10, that cutoff is 100. In base 2, it is 4, written 100_2.

The biggest two-digit number is one less than that cutoff, $b^2 - 1$. (In base 10 it is 99.) So you need to show the worst sum is no bigger than $b^2 - 1$. Start from that inequality and move everything to one side:

$3(b-1) \le b^2 - 1$	what we want
$3b - 3 \le b^2 - 1$	expand
$0 \le b^2 - 3b + 2$	subtract $3b - 3$ from both sides
$0 \le (b-1)(b-2)$	factor.

When $b \ge 2$, the factor $b-1$ is positive and the factor $b-2$ is zero or positive, so their product is never negative. The last line is true, and each step can be reversed, so the first line is true for every integer $b \ge 2$.

Check with base 10: $3 \cdot 9 = 27 \le 99$, and $(10-1)(10-2) = 72 \ge 0$.

Base 2 is the tight case. $1 + 1 + 1 = 3$, written 11_2. Two digits, just barely.

1.2 Binary length versus decimal length

The number 13 is two decimal digits and four binary digits: 1101_2. The question is whether binary is ever more than four times as long.

The number of digits of N in base b is about $\log_b N$. Changing base is multiplication by a constant:

$$\log_2 N = \log_2 10 \cdot \log_{10} N \approx 3.322 \log_{10} N.$$

So for a huge number, binary should be about 3.322 times as long as decimal. That is under 4. The rest of the proof is making sure the rounding never pushes a real number over 4. **One decimal digit.** The numbers are 1 through 9. The longest in binary is $9 = 1001_2$, four bits. Four times one digit. Fine.

Two or more decimal digits. Say N has d decimal digits, with $d \geq 2$. The exact digit count in binary is "round $\log_2 N$ down, then add 1," which is at most $\log_2 N + 1$. And a d-digit number is less than 10^d (a two-digit number is less than 100). Chain these together:

$$\begin{aligned}
\text{bits} &\leq \log_2 N + 1 && \text{rounding down only helps} \\
&< \log_2 \left(10^d\right) + 1 && \text{because } N < 10^d \\
&= d \log_2 10 + 1 && \text{pull the exponent out} \\
&< 3.33\, d + 1 && \text{because } \log_2 10 \approx 3.322 \\
&\leq 3.33\, d + 0.5\, d && \text{because } d \geq 2 \text{ means } 1 \leq d/2 \\
&= 3.83\, d \\
&< 4d.
\end{aligned}$$

So the binary length is under four times the decimal length.

Check with $N = 99$: it has $d = 2$ decimal digits, and $99 = 1100011_2$ has 7 bits. $7 \leq 8$.

The second sentence of the problem is the 3.322. Write $\log_2 10$, or about 3.32.

1.3 Why a tree cannot be both narrow and short

Tools used: T4, T6, T8, T13 and T14.

A d-ary tree means each person in a family tree has at most d children. Depth means how many steps you are from the root. The root has depth 0. Its children have depth 1.

Suppose the depth is only D. How many people can there be?

- Level 0: 1 person.
- Level 1: at most d.
- Level 2: at most d^2.
- Level D: at most d^D.

Add those up. It is the same kind of sum as exercise 0.2, with $c = d$ and last exponent D, so the shortcut formula applies:

$$n \leq 1 + d + d^2 + \cdots + d^D = \frac{d^{D+1} - 1}{d - 1}.$$

Make the right side simpler. Dropping the -1 on top makes it bigger, and since $d \geq 2$ the bottom $d - 1$ is at least 1, so removing it makes it bigger again (or leaves it the same):

$$\frac{d^{D+1} - 1}{d - 1} < \frac{d^{D+1}}{d - 1} \leq d^{D+1}.$$

So if the tree has n nodes,

$$n < d^{D+1}.$$

Now solve for D by taking a log of both sides:

$$\begin{aligned}
\log n &< \log\left(d^{D+1}\right) && \\
\log n &< (D+1)\log d && \text{pull the exponent out} \\
\frac{\log n}{\log d} &< D+1 && \text{divide by } \log d \\
D &> \frac{\log n}{\log d} - 1. &&
\end{aligned}$$

A tree with a child limit cannot stay short. The depth has to be at least about $\log n / \log d$. That is the Ω claim.

The exact minimum. The shortest possible tree is the one you fill level by level, with no gaps. It can hold n nodes as soon as the full count reaches n. Solve that for D:

$$\begin{aligned}
\frac{d^{D+1}-1}{d-1} &\geq n && \text{enough room for } n \text{ nodes} \\
d^{D+1} - 1 &\geq n(d-1) && \text{multiply by } d-1 \\
d^{D+1} &\geq 1 + n(d-1) && \text{add } 1 \\
D+1 &\geq \log_d\big(1+n(d-1)\big) && \text{take } \log_d \\
D &\geq \log_d\big(1+n(d-1)\big) - 1. &&
\end{aligned}$$

D has to be a whole number, so take the smallest whole number that works:

$$D = \left\lceil \log_d\big(1+n(d-1)\big) \right\rceil - 1.$$

The ceiling means "round up to a whole number of levels."

Check with $d = 2$ and $n = 7$: $\log_2(1 + 7 \cdot 1) = \log_2 8 = 3$, so $D = 3 - 1 = 2$. A full binary tree with levels of 1, 2 and 4 nodes has 7 nodes and depth 2.

If $d = 1$, nobody has two children, the tree is a straight line, and the depth is $n - 1$. The log formula needs $d \geq 2$.

1.4 and 1.5 A product and a slow sum

Tools used: T4, T6, T8, T9 and T16.

1.4 $\log(n!)$ is the same speed as $n \log n$

$n!$ means $1 \cdot 2 \cdot 3 \cdots n$. Taking a log turns multiplication into addition, which is why the problem asks about $\log(n!)$ instead of $n!$ itself.

Upper bound. Every factor is at most n, and there are n factors:

$$n! = 1 \cdot 2 \cdot 3 \cdots n \leq \underbrace{n \cdot n \cdot n \cdots n}_{n \text{ factors}} = n^n.$$

Take the log of both sides and pull the exponent out:

$$\log(n!) \le \log\left(n^n\right) = n \log n.$$

That is $O(n \log n)$.

Lower bound. Throw away the small half of the factors. The big half, about $n/2$ numbers, are all at least $n/2$. For example, with $n = 8$:

$$8! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 \cdot 8 \ge 5 \cdot 6 \cdot 7 \cdot 8 \ge 4 \cdot 4 \cdot 4 \cdot 4 = 4^4.$$

In general,

$$n! \ge \left(\frac{n}{2}\right)^{n/2}.$$

Take the log of both sides, pull the exponent out, then split the log of the fraction:

$$\begin{aligned}
\log(n!) &\ge \log\left(\left(\frac{n}{2}\right)^{n/2}\right) \\
&= \frac{n}{2} \log \frac{n}{2} && \text{pull the exponent out} \\
&= \frac{n}{2}(\log n - \log 2) && \log(a/b) = \log a - \log b \\
&= \frac{1}{2}\, n \log n - \frac{\log 2}{2}\, n && \text{expand.}
\end{aligned}$$

The first piece is half of $n \log n$. The second piece is only a constant times n, which is slower than $n \log n$, so subtracting it does not change the speed. To confirm, divide by $n \log n$:

$$\frac{\frac{1}{2}\, n \log n - \frac{\log 2}{2}\, n}{n \log n} = \frac{1}{2} - \frac{\log 2}{2 \log n} \to \frac{1}{2}.$$

A positive constant, so the lower bound is the same speed as $n \log n$. Together, the two bounds say $\Theta(n \log n)$.

You do not need a fancy formula for $n!$. The hint is the whole proof.

1.5 $1 + 1/2 + 1/3 + \cdots + 1/n$

This sum grows, but slowly. The problem mentions $\ln n + 0.577$, which is a sharper estimate. You do not need that number. You only need to show the sum is the same speed as $\log n$.

The trick is to group the fractions into blocks that start or end at powers of 2.

Upper bound. Start each block at a power of two:

$$1 + \left(\frac{1}{2} + \frac{1}{3}\right) + \left(\frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7}\right) + \cdots$$

In each block, replace every denominator by the power of two at the start of the block. A smaller denominator makes a bigger fraction, so this overestimates the sum:

$$\begin{gathered}
\frac{1}{2} + \frac{1}{3} \le \frac{1}{2} + \frac{1}{2} = 2 \cdot \frac{1}{2} = 1, \\
\frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} \le \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = 4 \cdot \frac{1}{4} = 1.
\end{gathered}$$

The pattern continues: the block starting at $1/2^k$ has 2^k fractions, each at most $1/2^k$, so it adds up to at most $2^k \cdot \frac{1}{2^k} = 1$. Every block is worth at most 1.

How many blocks are there? They start at $1, 2, 4, 8, \ldots$, so the number of blocks needed to reach n is the number of powers of two up to n, which is at most $\log_2 n + 1$. So

$$1 + \frac{1}{2} + \cdots + \frac{1}{n} \leq \underbrace{1 + 1 + \cdots + 1}_{\text{at most } \log_2 n + 1 \text{ blocks}} \leq \log_2 n + 1.$$

Lower bound. Do the opposite. End each block at a power of two:

$$1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \cdots$$

Replace every denominator by the power of two at the end of the block. A bigger denominator makes a smaller fraction, so this underestimates the sum:

$$\frac{1}{3} + \frac{1}{4} \geq \frac{1}{4} + \frac{1}{4} = 2 \cdot \frac{1}{4} = \frac{1}{2},$$
$$\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} \geq \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = 4 \cdot \frac{1}{8} = \frac{1}{2}.$$

The pattern continues: the block ending at $1/2^k$ has 2^{k-1} fractions, each at least $1/2^k$, so it adds up to at least $2^{k-1} \cdot \frac{1}{2^k} = \frac{1}{2}$. Every full block is worth at least $1/2$.

The number of full blocks that fit below n is about $\log_2 n$ (at least $\log_2 n - 1$). So

$$1 + \frac{1}{2} + \cdots + \frac{1}{n} \geq 1 + \frac{1}{2}(\log_2 n - 1) = \frac{1}{2}\log_2 n + \frac{1}{2}.$$

Check with $n = 8$: the sum is about 2.72. The lower bound gives $\frac{1}{2} \cdot 3 + \frac{1}{2} = 2$ and the upper bound gives $3 + 1 = 4$. And $2 \leq 2.72 \leq 4$.

Trapped above and below by constant multiples of $\log n$. That is $\Theta(\log n)$.

A PICTURE OF THE TRICK

You are not computing the sum. You are trapping it. Make every term a little too big, and the blocks become easy. Make every term a little too small, and the blocks become easy again. If both easy sums are "about $\log n$," the real sum is too.

1.9 and 1.10 Remainders

Tools used: T5 and T15.

$x \equiv y \pmod{N}$ means x and y leave the same remainder when divided by N. The form the problem wants is:

$$x \equiv y \pmod{N} \quad \text{means} \quad N \text{ divides } x - y.$$

Example: $17 \equiv 2 \pmod{5}$, because $17 - 2 = 15$, and 5 divides 15. Equivalently, $17 = 2 + 3 \cdot 5$.

1.9 You can swap in a remainder before adding or multiplying

Suppose x and x' leave the same remainder, and so do y and y'. Write

$$x = x' + aN, \qquad y = y' + bN.$$

Here a and b are whole numbers: they count how many extra copies of N each one carries. (For $17 = 2 + 3 \cdot 5$, that count is 3.)

Adding. Substitute both, then collect the pieces with an N:

$$\begin{aligned} x + y &= (x' + aN) + (y' + bN) && \text{substitute} \\ &= x' + y' + aN + bN && \text{reorder} \\ &= (x' + y') + (a + b)N && \text{factor out } N. \end{aligned}$$

Move $x' + y'$ to the left:

$$(x + y) - (x' + y') = (a + b)N.$$

The difference is a whole number times N, which is exactly what $x + y \equiv x' + y' \pmod{N}$ means.

Try it: $17 + 9 = 26$, and $2 + 4 = 6$. Both leave remainder 1 when divided by 5.

Multiplying. Substitute both and multiply out the brackets, each term in the first times each term in the second:

$$\begin{aligned} xy &= (x' + aN)(y' + bN) && \text{substitute} \\ &= x'y' + x'bN + aNy' + aN \cdot bN && \text{multiply out} \\ &= x'y' + N(x'b + ay' + abN) && \text{factor } N \text{ out of the last three.} \end{aligned}$$

Move $x'y'$ to the left:

$$xy - x'y' = N(x'b + ay' + abN).$$

The bracket is a whole number, so the difference is again a whole number times N. That means $xy \equiv x'y' \pmod{N}$.

Try it: $17 \cdot 9 = 153$, and $2 \cdot 4 = 8$. Both leave remainder 3 when divided by 5, because $153 = 30 \cdot 5 + 3$.

This is why, on a clock, you can reduce as you go. Adding 17 hours is the same as adding 5 hours on a 12-hour clock.

1.10 A divisor of the clock size still works

If $a - b$ is a multiple of N, and M divides N, then $a - b$ is also a multiple of M.

Numbers: $26 \equiv 2 \pmod{12}$, because the difference is 24. And 4 divides 12, so $26 \equiv 2 \pmod{4}$ as well.

In symbols, write down what each fact means, then substitute one into the other:

$$\begin{aligned} a - b &= kN && a \equiv b \pmod{N},\ k \text{ a whole number} \\ N &= mM && M \text{ divides } N,\ m \text{ a whole number} \\ a - b &= k(mM) && \text{substitute for } N \\ a - b &= (km)M && \text{regroup.} \end{aligned}$$

km is a whole number, so $a - b$ is a whole number times M. That is what $a \equiv b \pmod{M}$ means.

With the numbers above: $26 - 2 = 24 = 2 \cdot 12$, so $k = 2$. And $12 = 3 \cdot 4$, so $m = 3$. Then $24 = (2 \cdot 3) \cdot 4 = 6 \cdot 4$.

DO NOT REVERSE IT

$5 \equiv 1 \pmod{4}$, but 5 and 1 do not leave the same remainder modulo 12. Going to a smaller divisor can only lose information. It cannot create a stronger statement.

How to write the answer

1. Say which claim you are proving, and for which n. "For all $n \geq 6$" is different from "for all n."
2. For a growth question, write the fraction and say whether it goes to a number, to 0, or to infinity.
3. For induction, write the two base cases, then one sentence that starts "Assume this is true for the previous two values."
4. For a remainder proof, rewrite $\equiv$ as "the difference equals an integer times N" before you do anything else.

Then compare with `solutions.tex`. If your argument has the same skeleton, the wording does not have to match.